

କେହାଟି ଐସିଆର ପ୍ରତି?

Here's a look at the Indian space agency's most recent achievements and what other record breaking feats we can expect in the future

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*"India becomes first Asian nation to reach Marsorbit, joins elite global space club".
"ISRO sets Record, Launches PSLV-C34 with 20 satellites". "At a stroke, PSLV C-34 lobs 20 satellites into orbit".*

these are just a few of the headlines from leading newspapers around the globe, praising ISRO for some of the many remarkable feats it has achieved in recent times.

From being the first space research agency of a newborn nation to becoming the first nation in the world to reach Mars in its first attempt, to launching 20 satellites (including 17 foreign payloads) in 26 minutes with its indigenously developed launcher, ISRO has come a long way since it's inception more than 50 years ago.

Recently, PSLV-C34 (Polar Satellite Launch Vehicle) was successfully launched from Satish Dhawan Space Centre on 22nd June this year. Carrying a total of 20 satellites, the primary payload of the launch vehicle was Cartosat-2C along with 19 other satellites including one for a Google company called Terra Bella and the rest from US, Canada, Germany and Indonesia. The first of its kind mission made this the 3rd biggest launch in the world, only behind NASA and the Russian Space Agency.

After successfully reaching the Moon and Mars, ISRO has its eyes set on its next set of targets. Read on to find out what ISRO has planned out for its future

WHAT THE FUTURE HOLDS?

From developing an indigenous scramjet engine that will let a launch vehicle carry more than 20 satellites, to landing a rover on Mars and the moon, and also sending another satellite to the nearby Venus, ISRO has lots of innovative plans lined up. Here is a comprehensive list of all the projects it has been working on including its latest launch vehicles, satellites and other space exploration programmes.

Launch vehicles:

- **Reusable Launch Vehicle** - Unlike normal launch vehicles which are expendable, a Reusable Launch Vehicle is, well, reusable and can be used to launch a payload into space more than once. The Reusable Launch Vehicle Technology demonstrator is a prototype being developed by ISRO which is going to be developed into a fully functional Launch Vehicle by the year 2030. The indigenously developed launch vehicle, designed by a team of 750 engineers, has already completed its first successful test flight on 23rd May where it

flew for a total of 770 seconds reaching a maximum altitude of 65 Kilometres before safely landing back on earth. The technology, if successful, will bring down the overall payload delivery cost drastically by almost 80%.

- **Scramjet Engine** - Unlike conventional launchers which carry liquid oxygen or an oxidiser to fire the engine, a scramjet engine inhales air from the atmosphere and uses its oxygen to burn the fuel. Use of scramjet engine leads to faster and cheaper rockets compared to the traditional liquid or cryogenic engines. ISRO has been working on developing its own scramjet engine for quite some time now, and a small model by them is going to be tested this month. Once developed, the engine will help put satellites and launch vehicles at significantly lower costs bringing down the overall cost of space exploration.
- **GSLV MK III/ LVM 3** - GSLV Mark 3 or more commonly known as the LVM 3, is a satellite launcher being developed by ISRO that will make India completely self-reliant in launching satellites, capable of placing 4 tonne class Geosynchronous satellites into orbit. Powered by an indigenously developed Cryogenic engine, the launcher will be used to launch var-